

Calculus 1 Problem Templates for Dynamic Generation

Master Reference Sheet for Generator Patterns

Calculus Practice System (v.2026.01)

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Instructions for Professor: This document collects representative problem archetypes across all 7 core categories in the Calculus 1 practice hub. Please write your step-by-step gold-standard solution and LaTeX layout inside the `solutionbox` for each problem type. Once completed, the system will use your exact structure as the universal dynamic template for all randomized problem instances.

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1 Limits of Functions

1.1 Type 1.1: Factoring (0/0 Quadratic / Difference of Squares)

Problem 1.1: Limit by Factoring

Evaluate the limit:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Gold-Standard Solution (Write your master solution template here)

1.2 Type 1.2: Square Root Conjugate (0/0)

Problem 1.2: Limit by Conjugate (Square Root)

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$$

Gold-Standard Solution (Write your master solution template here)

1.3 Type 1.3: Cube Root Conjugate / Difference of Cubes (0/0)

Problem 1.3: Limit by Cube Root Conjugate

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{x+8} - 2}{x}$$

Gold-Standard Solution (Write your master solution template here)

1.4 Type 1.4: Complex Fraction (0/0)

Problem 1.4: Limit of Complex Fractions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\frac{1}{x+3} - \frac{1}{3}}{x}$$

Gold-Standard Solution (Write your master solution template here)

1.5 Type 1.5: Double Conjugate (Numerator & Denominator)

Problem 1.5: Double Conjugate Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{\sqrt{x+9} - 3}$$

Gold-Standard Solution (Write your master solution template here)

1.6 Type 1.6: Basic Trigonometric Limits ($\frac{\sin kx}{x}$, $\frac{\tan kx}{\sin mx}$)

Problem 1.6: Trigonometric Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(5x)}$$

Gold-Standard Solution (Write your master solution template here)

1.7 Type 1.7: Limits at Infinity (∞/∞ & $\infty - \infty$)

Problem 1.7: Limit at Infinity

Evaluate the limit:

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 4x} - x)$$

Gold-Standard Solution (Write your master solution template here)

1.8 Type 1.8: One-Sided Limit with Absolute Value

Problem 1.8: One-Sided Limit with $|x|$

Evaluate the one-sided limit:

$$\lim_{x \rightarrow 2^-} \frac{|x-2|}{x^2-4}$$

Gold-Standard Solution (Write your master solution template here)

1.9 Type 1.9: L'Hôpital's Rule ($0/0$ or ∞/∞)

Problem 1.9: L'Hôpital's Rule

Evaluate the limit using L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin(3x)}$$

Gold-Standard Solution (Write your master solution template here)

2 Basic Derivatives & Term-by-Term Differentiation

2.1 Type 2.1: Polynomial with Fractional and Negative Powers

Problem 2.1: Powers and Radicals

Given $f(x) = 3x^3 + \frac{3}{x} + 5\sqrt{x}$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}u^n &= \frac{1}{u^{-n}} \\ \sqrt[n]{u^m} &= x^{\frac{m}{n}} \\ [u^n]' &= nu^{n-1}u' \\ (k_1f \pm k_2g)' &= k_1f' \pm k_2g'\end{aligned}$$

Step 2. Then,

$$f'(x) = \frac{d}{dx} \left(3x^3 + \frac{3}{x} + 5\sqrt{x} \right) = 3 \frac{d}{dx} x^3 + 3 \frac{d}{dx} \frac{1}{x} + 5 \frac{d}{dx} \sqrt{x}. \quad (1)$$

Consider 1st term, then

$$\frac{d}{dx} x^3 = 3x^2 \frac{d}{dx} x = 3x^2(1) = 3x^2.$$

Consider 2nd term, then

$$\frac{d}{dx} \frac{1}{x} = \frac{d}{dx} x^{-1} = (-1)x^{-2} \frac{d}{dx} x = (-1)x^{-2}(1) = -\frac{1}{x^2}.$$

Consider 3rd term, then

$$\frac{d}{dx} \sqrt{x} = \frac{d}{dx} x^{\frac{1}{2}} = \frac{1}{2} \cdot x^{-\frac{1}{2}} \frac{d}{dx} x = \frac{1}{2} \cdot x^{-\frac{1}{2}}(1) = \frac{1}{2x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}.$$

Combine these three terms together, we obtain

$$\begin{aligned}f'(x) &= 3 \frac{d}{dx} x^3 + 3 \frac{d}{dx} \frac{1}{x} + 5 \frac{d}{dx} \sqrt{x} \\ &= 3(3x^2) + 3\left(-\frac{1}{x^2}\right) + 5\left(\frac{1}{2\sqrt{x}}\right) \\ &= 9x^2 - \frac{3}{x^2} + \frac{5}{2\sqrt{x}}\end{aligned}$$

2.2 Type 2.2: Basic Trigonometric Functions (Sine, Cosine, Secant, Tangent)

Problem 2.2: Trig Derivatives

Given $f(x) = 4\sin(3x) - 2\cos(3x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

2.3 Type 2.3: Natural and General Exponential Functions (e^{kx} and a^u)

Problem 2.3: Exponential Derivatives

Given $f(x) = 3^{2x+1} + 2e^{3x}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

2.4 Type 2.4: Sum of Inverse Trigonometric Functions

Problem 2.4: Inverse Trig Derivatives

Given $f(x) = 3 \arccos(4x) + 2 \operatorname{arccot}(x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

3 Product Rule: $(uv)' = uv' + vu'$

3.1 Type 3.1: Polynomial $\times$ Exponential

Problem 3.1: Product Rule (Poly $\times$ Exp)

Given $f(x) = (2x + 3)e^{2x}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

3.2 Type 3.2: Exponential $\times$ Trigonometric (Damped Oscillation)

Problem 3.2: Product Rule (Exp $\times$ Trig)

Given $f(x) = e^{-2x} \cos(3x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

3.3 Type 3.3: Polynomial / Exponential $\times$ Inverse Trigonometric

Problem 3.3: Product Rule with Inverse Trig

Given $f(x) = (e^{3x} + 2x) \arctan(x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

4 Quotient Rule: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

4.1 Type 4.1: Rational Polynomial Function

Problem 4.1: Quotient Rule (Polynomials)

Given $f(x) = \frac{2x+3}{x-4}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

4.2 Type 4.2: Trigonometric / Radical Quotient

Problem 4.2: Quotient Rule (Trig / Radical)

Given $f(x) = \frac{\sin(2x)}{\sqrt{x^2+1}}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

5 Chain Rule & Composite Functions

5.1 Type 5.1: Radical with Quadratic Inside $\sqrt{u(x)}$

Problem 5.1: Chain Rule with Radical

Given $f(x) = \sqrt{2x^2 + 3x + 4}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

5.2 Type 5.2: Logarithmic with Polynomial Inside $\ln(u(x))$

Problem 5.2: Chain Rule with Natural Log

Given $f(x) = \ln(x^2 + 3x + 2)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

5.3 Type 5.3: Trigonometric with Linear / Quadratic Inside

Problem 5.3: Chain Rule with Cotangent

Given $f(x) = 2 \cot\left(3x + \frac{\pi}{4}\right)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

5.4 Type 5.4: Inverse Trigonometric with Chain Rule

Problem 5.4: Chain Rule with ArcSec

Given $f(x) = 2 \operatorname{arcsec}(3x)$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

5.5 Type 5.5: Nested Composite Function (Trig(Inverse Trig(kx)))

Problem 5.5: Nested Trig of Inverse Trig

Given $f(x) = \tan(\arcsin(3x))$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

5.6 Type 5.6: Logarithmic Differentiation ($y = x^{kx}$)

Problem 5.6: Logarithmic Differentiation

Given $f(x) = x^{3x}$ for $x > 0$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

5.7 Type 5.7: Higher-Order Derivative ($f''(x)$)

Problem 5.7: Second Derivative

Given $f(x) = x^2 e^{2x}$, find $f''(x)$ and evaluate $f''(0)$.

Gold-Standard Solution (Write your master solution template here)

6 Implicit Differentiation

6.1 Type 6.1: Circle / Quadratic Relation

Problem 6.1: Implicit Differentiation (Circle)

Find $\frac{dy}{dx}$ for the equation $x^2 + y^2 = 25$ at the point $(3, 4)$.

Gold-Standard Solution (Write your master solution template here)

6.2 Type 6.2: Mixed Polynomial with xy Product Term

Problem 6.2: Implicit Differentiation with Product Rule

Find $\frac{dy}{dx}$ for the equation $x^3 + y^3 - 3xy = 5$ at the point (x_0, y_0) .

Gold-Standard Solution (Write your master solution template here)

7 Tangent & Normal Lines

7.1 Type 7.1: Equation of the Tangent Line

Problem 7.1: Tangent Line Equation

Find the slope of the tangent line m_t and the equation of the tangent line to the curve $y = x^2 - 3x + 2$ at $x = 2$.

Gold-Standard Solution (Write your master solution template here)

7.2 Type 7.2: Equation of the Normal Line

Problem 7.2: Normal Line Equation

Find the slope of the normal line m_n and the equation of the normal line to the curve $y = x^2 - 3x + 2$ at $x = 2$.

Gold-Standard Solution (Write your master solution template here)